EE5606/AI5120 : Convex Optimization

Robust Convex Formulations for Sparse Recovery under Heavy-Tailed Noise (RADAR example)

TEAM 22:

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భారతీయ సాంకేతిక విజ్ఞాన మరియు హైదరాబాద్
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Introduction:

- Millimeter-wave (mmWave) radar is a highly reliable sensor for autonomous systems, unaffected by extreme weather or poor lighting.
- However, mmWave radar inherently produces highly **sparse data** because physical surfaces often reflect radar waves away from the sensor.
- To extract useful 3D geometric information, we must effectively reconstruct the missing structure from this extremely **sparse data**.
- We are formulating the process of recovering this sparse data as a **optimization problem** particularly using **Compressed Sensing** as the radar outputs sparser data.
- We will solve for this optimization problem to get an optimal information like position/ AoA of object from corrupted reflections.

Motivation:

- Traditional Fast Fourier Transform (FFT) techniques suffer from physical resolution limits and blur closely spaced targets together.
- To achieve super-resolution, we replace FFT with **compressed sensing**, which perfectly recovers targets by solving an underdetermined linear equation.
- In **compressed sensing**, recovering the signal is framed as a **convex optimization problem** to find the sparsest possible explanation for the antenna measurements.
- However, indoor radar environments suffer from severe multipath interference, generating massive "ghost" echoes that act as **heavy tailed noise**.
- Standard algorithms use quadratic **loss functions** (like l2) that heavily penalize large errors and only performs for Gaussian noise, causing the entire optimization to fail and warp the point cloud when hit by this **heavy tailed noise**. Therefore we replace that with Huber loss which can deal with all types of noises.

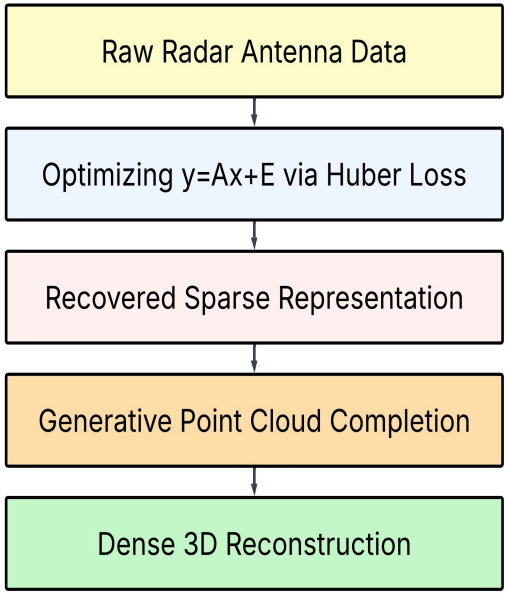
Literature Survey:

Reference	Algorithm Type	Primary Loss Function	Target Noise Profile	Complexity	Key Limitation / Relevance
Bui (2025)	Quantum FFT (QFT)	N/A (Linear Transform)	Thermal Gaussian	O(N log N)	Fast, but fundamentally limited by Rayleigh resolution and blurring under multipath.
Mohammad (2025)	CS (LASSO / Greedy)	L2 (Standard) vs L1 (Robust)	Gaussian vs Outliers	$O(N^3)$	Standard LASSO fails under impulsive errors; establishes L1 viability for robustness.
Zheng (2026)	Approx. Message Passing	L2 Fidelity (Gaussian)	Standard Gaussian	O(MN)	Fast CS solver; breaks Rayleigh limit but warps geometries when hit by multipath outliers.
Mlotshwa (2022)	Robust Regression	Cauchy Loss	Cauchy / Impulsive	$O(N^2)$	Proves non-quadratic penalties are required to handle massive outliers effectively.

Mecklenbräuker (2024)	Robust M-Estimation	Huber / Tyler Penalty	Heavy-Tailed Radar	$O(N^2)$	Validates Huber for array processing, mathematically filtering out "ghost" DOA signals.
Chen (2025)	Deep Invertible Diffusion	Learned Score Matching	Complex Non-Gaussian	Very High	Structurally completes missing geometric data but requires pristine, denoised inputs.
Proposed Method	**Huber-AMP + AI Gen**	**Huber Fidelity + L1**	**Cauchy / Multipath**	**O(MN)**	**Combines fast AMP solver with robust Huber outlier rejection for clean generative completion.**

Methodology:

- We propose a two-stage "Robust-Generative" pipeline to solve the limitations of standard radar recovery.
- **Stage 1 (Robust Front-End):** We modify the classical convex optimization problem by replacing the standard quadratic loss functions with an adaptive Huber loss.
- This creates a "Huber-AMP" algorithm that puts a hard limit on penalties, mathematically ignoring the massive outliers caused by heavy tailed noise.
- **Stage 2 (Generative Back-End):** The clean, accurately located sparse data output from Huber-AMP is cascaded into a deep generative neural network.
- The neural network acts as a structural prior, successfully hallucinating the missing geometric surfaces lost to specular reflection.



To represent the physical radar sensing process, we use the standard underdetermined linear equation:

- **Equation: y = Ax + epsilon**
  - **y (Measurements):** The raw electrical signals recorded by the radar's antennas.
  - **A (Sensing Matrix):** The physical "rulebook" modeling the phase delays between antennas.
  - **x (The Truth):** The sparse spatial spectrum (the actual angles of the targets).
  - **epsilon (The Noise):** Environmental noise, including massive multipath "ghost" reflections.
- The objective function :

$$\textbf{Standard Objective: } \hat{x} = \arg\min_x \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|x\|_1$$

$$\textbf{Proposed Objective: } \hat{x} = \arg\min_x \sum_{i=1}^{M} \mathcal{H}_\tau(y_i - a_i^T x) + \lambda\|x\|_1$$

- The Huber loss is defined piecewise. For an error e:
  - If $|e| \leq \tau$: It uses $\frac{1}{2}|e|^2$ (Quadratic / ℓ_2).
  - If $|e| > \tau$: It uses $\tau|e| - \frac{1}{2}\tau^2$ (Linear / ℓ_1).

Huber- AMP

$$\psi(r) = \begin{cases} r & |r| \leq \delta \\ \delta \cdot \text{sign}(r) & |r| > \delta \end{cases}$$

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Experimentation and Results:

- For testing purpose we used Shapenet LiDAR data simulate radar kind of data.
- Our error analysis demonstrates that standard l2-based **compressed sensing** fails in error performance under these chaotic conditions, generating massive false artifacts.
- In contrast, our Huber-AMP formulation maintains a drastically lower Mean Squared Error (MSE) and higher RSNR, successfully rejecting the interference.

Optimization Method	**MSE (↓)**	**NMSE (↓)**	**RSNR (↑)**	**F1-Score (↑)**
Standard *l2* (Baseline)	0.6925	8.6006	-9.3453	0.1775
Standard *l1*	0.0753	0.9350	0.2919	**0.2588**
Standard Huber	0.0809	1.0052	-0.0225	0.2439
Huber-AMP (Proposed)	**0.0742**	**0.9219**	**0.3532**	0.1789*

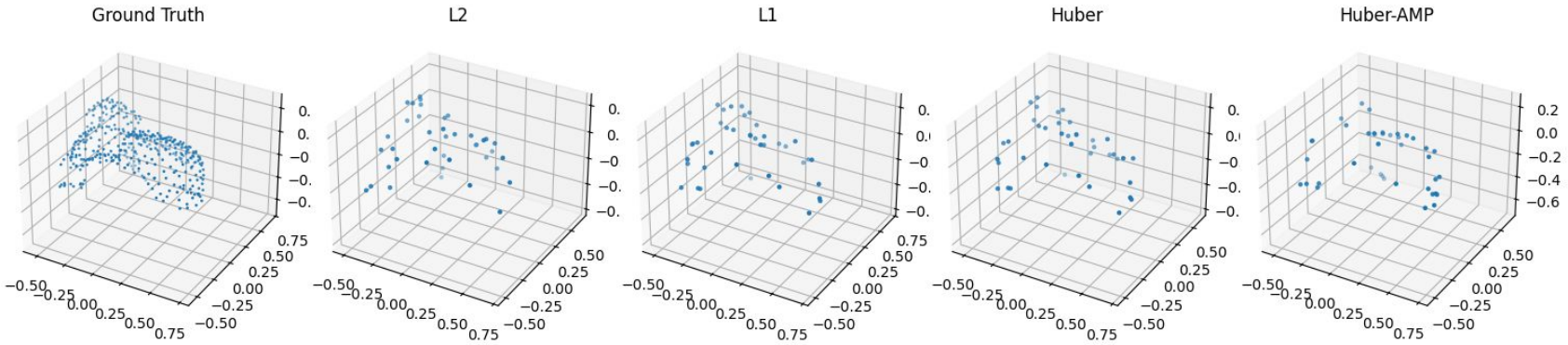


Fig: Comparison of all losses

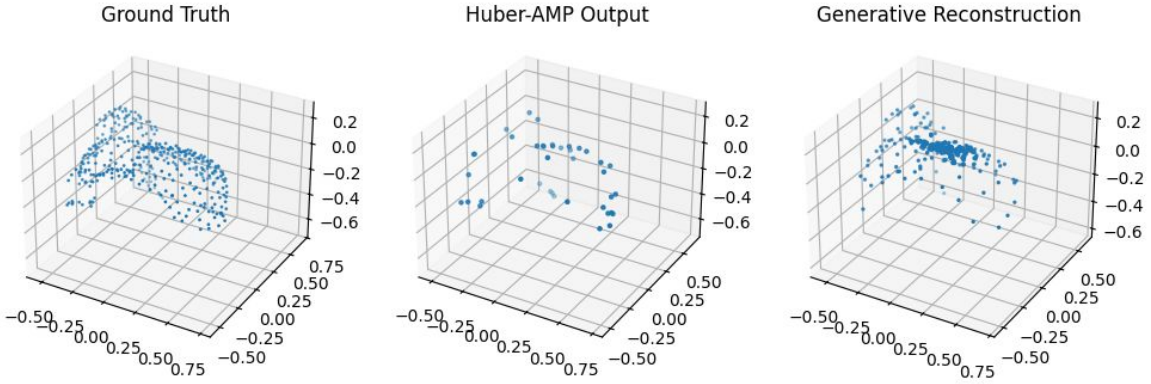


Fig: Generative Reconstruction(VAE)

Conclusion:

- We simulated raw radar measurements (y=Ax+e) on standard 3D point cloud datasets and injected severe Cauchy-distributed outliers.
- Our error analysis demonstrates that standard *l2*-based **compressed sensing** is over performed by Huber-AMP based loss in various performance parameters.
- In contrast, our Huber-AMP formulation maintains a drastically lower Mean Squared Error (MSE), successfully rejecting the interference.
- This proves our robust mathematical front-end provides the ideal, ghost-free foundation for generative networks to reconstruct the final dense 3D geometry.
- Also generative reconstruction after recovering clean sparse data using huber loss function was identical to the ground truth data,

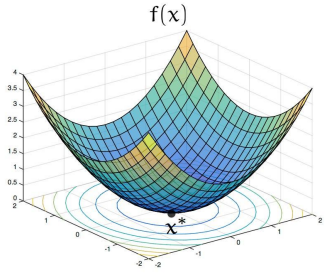
Individual Contributions

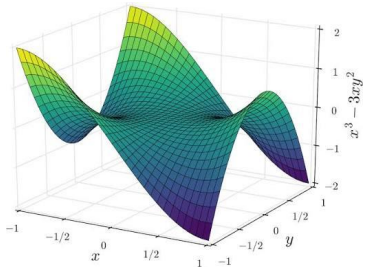
"*The strength of the team is each individual member. The strength of each member is the team.*" - Phil Jackson

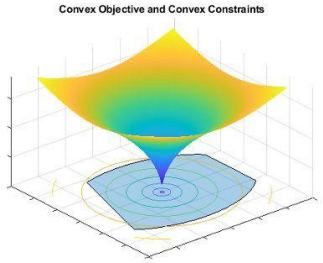
Inspired by the wisdom of above quote, we have divided our work equally among team members as follows:

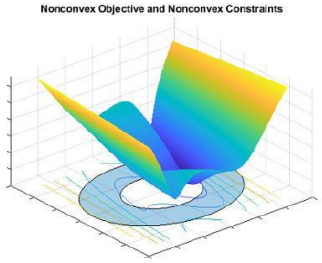
- **Veleti Karthikeya**: Problem formulation and Literature survey
- **Nikith**: Optimization instead of using standard FFT for signal recovery
- **Hemanth :** Survey on various Loss functions
- **Vignesh** : Results and Metric comparison
- **Sai Kiran** : Coding and execution part

The literature survey and code implementation were carried out individually by each member based on their assigned tasks.


f(x)
x*


$x^3 - 3xy^2$
x
y


Convex Objective and Convex Constraints


Nonconvex Objective and Nonconvex Constraints

Thank you